

CPSC 483: Introduction to Machine Learning

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Cal State Fullerton

Taxonomy of Machine Learning

What this course will focus on the most :)



Learning can be divided into broad categories, based on the following criteria:

- Whether or not they are trained with human supervision
 - i.e., supervised, unsupervised, semi-supervised, and Reinforcement Learning
- Whether or not they can learn incrementally on new data
 - i.e., online versus batch learning
- Whether they work by simply comparing new data points to known data points, or instead by detecting patterns in the training data and building a predictive model, much like scientists do
 - i.e., instance-based versus model-based learning

These criteria are not exclusive

Supervised Learning - Example of Data

Input	Output
Laura	F
Carlos	M
Jose	M
Maria	F
Belen	F



Iris Setosa



Iris Versicolor



Iris Virginica

Input

- The data or information given to an algorithm so it can analyze and learn

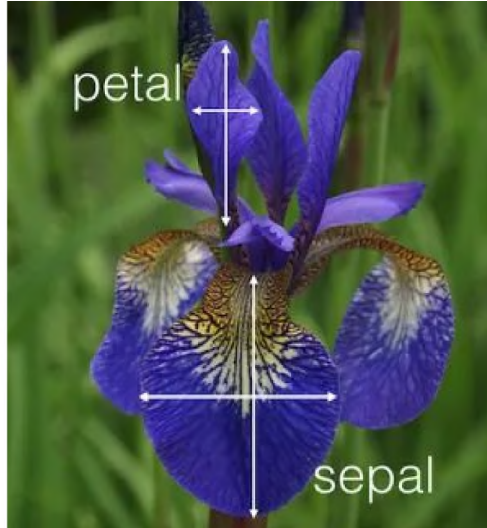
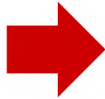
Output

- The final result, score, or prediction generated by the model after processing the input

Example of Data (cont'd)

Using Iris Flower Dataset, available:

<https://www.kaggle.com/datasets/arshid/iris-flower-dataset>



Input	nchars	lastchar	nvowels	Output
Laura	5	'a'	3	F
Carlos	6	's'	2	M
Jose	4	'e'	2	M
Maria	5	'a'	3	F
Belen	5	'n'	2	F

Feature/Feature Vectors

Label/Class

Sepal. Length	Sepal. Width	Petal. Length	Petal. Width	Output
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
7.0	3.2	4.7	1.4	versicolor
6.3	3.3	6.0	2.5	virginica
...	...	...	...	...

Supervised Learning - Structure of Dataset

Feature/Feature Vectors				Label/Class
Petal Length	Petal Width	Sepal Length	Sepal Width	Species
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
6.2	2.9	4.3	1.3	Versicolor
6.3	3.3	6.0	2.5	Virginica
5.1	2.5	3.0	1.1	Versicolor
7.1	3.0	5.9	2.1	Virginica
5.1	3.5	1.4	0.2	?
7.1	3.0	5.9	2.1	?

Dataset { *an instance*

Predictions {

Supervised Learning - Structure of Dataset (cont'd)

Feature/Feature Vectors				Label/Class
Petal Length	Petal Width	Sepal Length	Sepal Width	Species
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
6.2	2.9	4.3	1.3	Versicolor
6.3	3.3	6.0	2.5	Virginica
5.1	2.5	3.0	1.1	Versicolor
7.1	3.0	5.9	2.1	Virginica

Training Data

Testing Data

Split for Evaluation

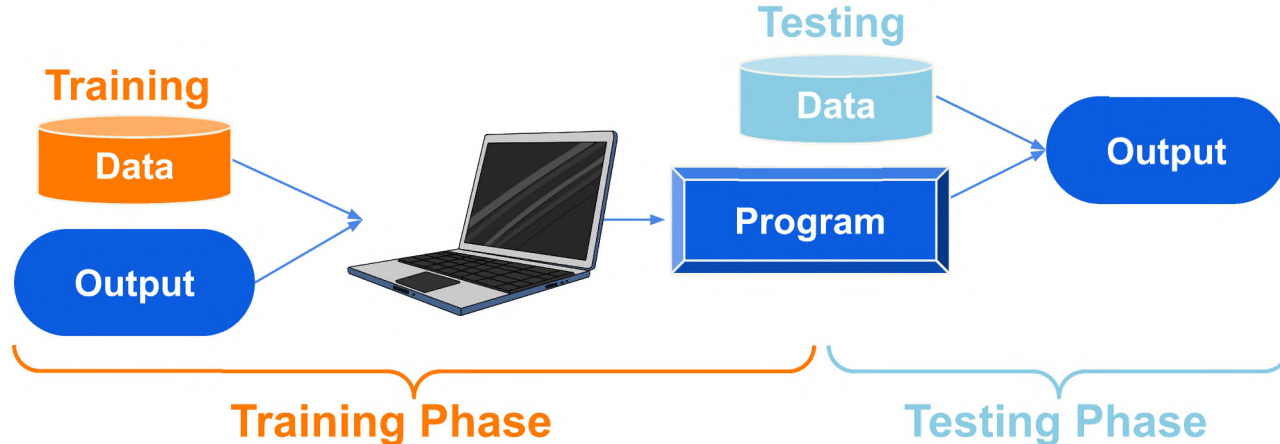
Supervised Learning - Structure of Dataset (*cont'd*)

Training Data

- Portion of data used to train/teach the machine learning model

Testing Data

- Portion of data used to test/evaluate the performance of a trained machine learning model



Supervised Learning - Structure of Dataset (cont'd)

Feature/Feature Vectors				Label/Class
Petal Length	Petal Width	Sepal Length	Sepal Width	Species
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	4.4	0.2	Setosa
6.2	2.9	5.5	1.3	Setosa
6.3	3.3	6.0	2.5	Virginica
5.1	2.5	5.0	1.1	Virginica
7.1	3.0	5.9	2.1	Virginica

Train Features (rows 1-3, columns 1-4)

Train Labels (rows 1-3, column 5)

Test Features (rows 4-6, columns 1-4)

Test Labels (rows 4-6, column 5)

Split for Evaluation

Note: Testing data should **NEVER!!!!** be used during training

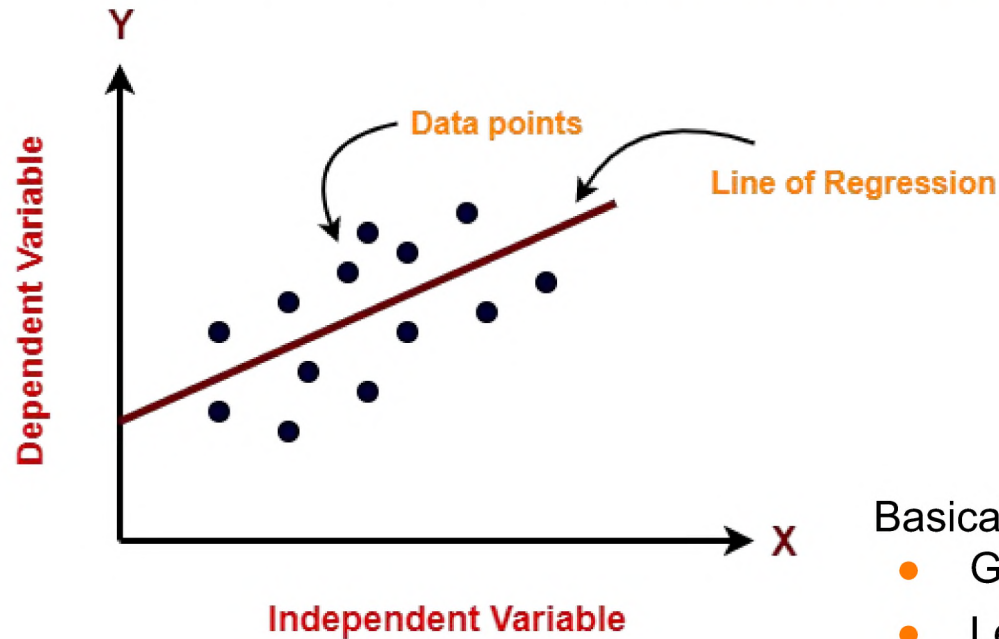
Supervised Learning - Definition

Given a **training dataset** $\{(X_1, \dots, X_k, Y_j) \mid j = 1 \dots N\}$

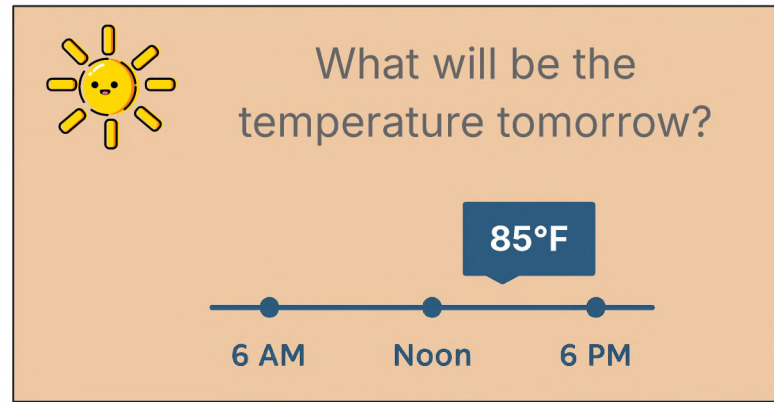
- Learn a **function** or **classifier** $f: X \rightarrow Y$
 - The process of learning f is called training
- Predict outcomes for a given X (new data):
 - If the learned function f is used to predict a continuous value, $\hat{y}$, it's called **regression**
 - If the learned classifier f is used to predict a discrete value, $\hat{y}$, it's called **classification**

Supervised Learning - *Regression*

- Identifies patterns to predict **continuous numerical** values



Example of Regression



Basically,

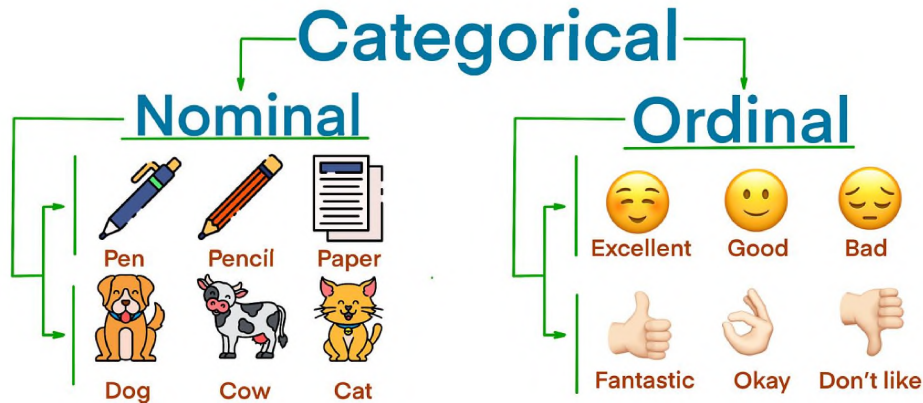
- Given $(x_1, y_1), (x_2, y_2), (x_3, y_3) \dots$
- Learn a **function** $f(x)$ to predict y given x

Supervised Learning - *Classification*

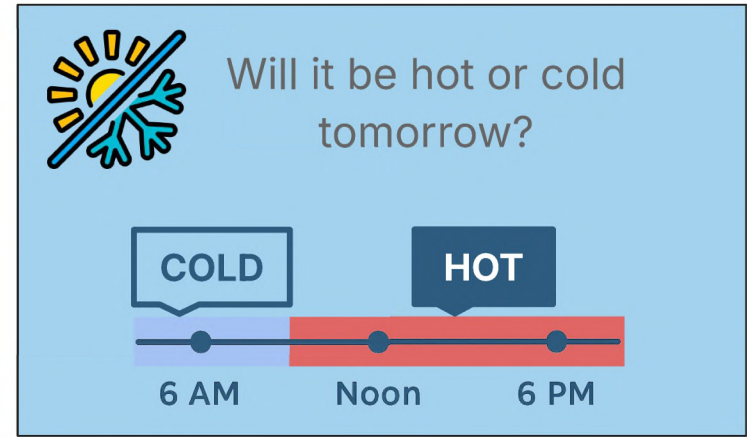
- Identifies patterns to predict **categorical** values



- Spam
- Not spam



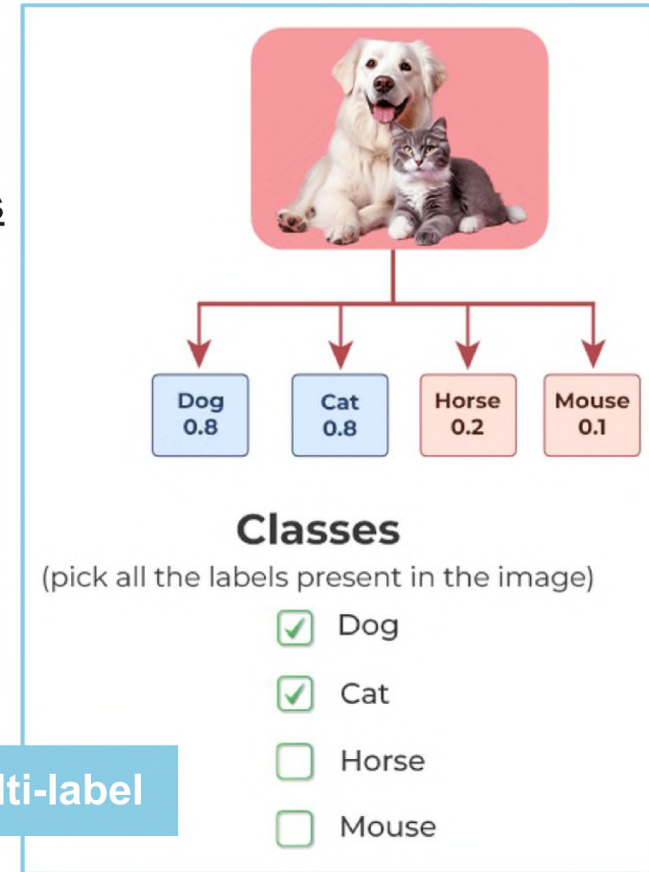
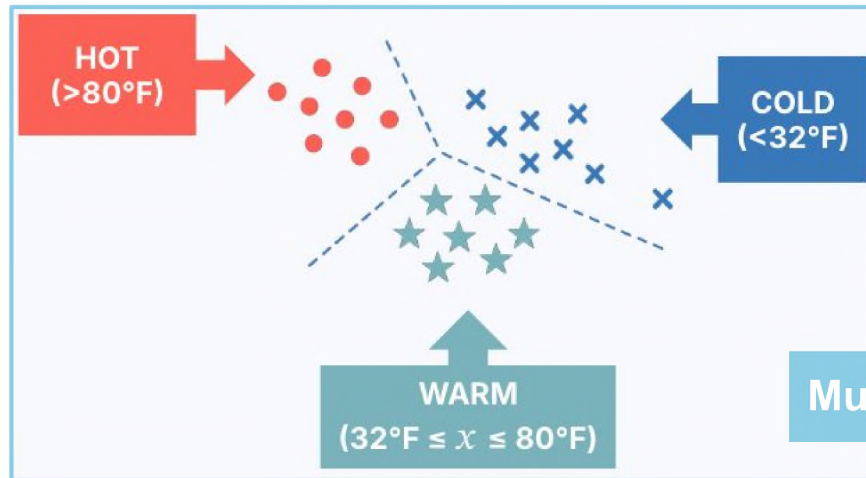
Example of Classification



Supervised Learning - *Classification*

x can be **multi-dimensional**

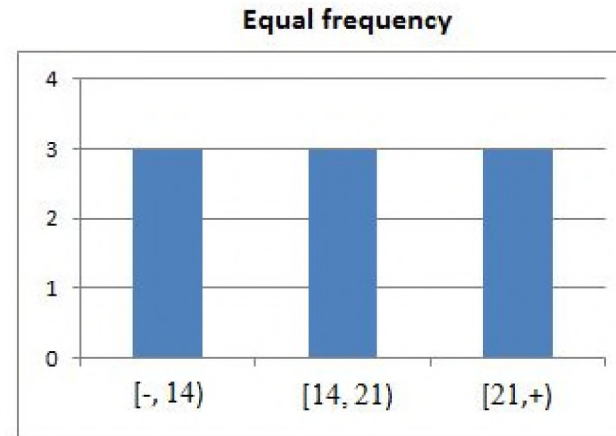
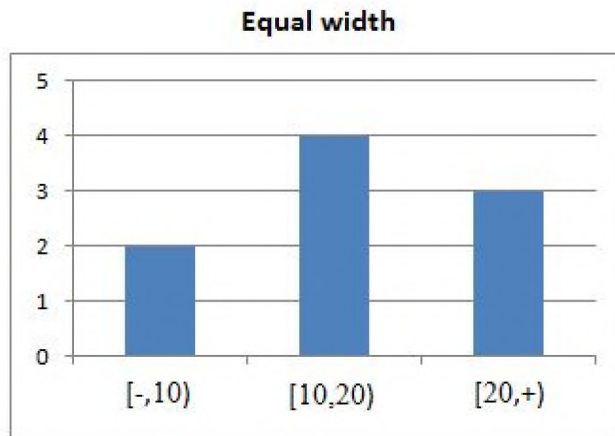
- **Multi-class**: each instance still belongs to only one class
 - but many possible classes to choose from
- **Multi-label**: each instance can have multiple labels at the same time



Supervised Learning - *Classification (cont'd)*

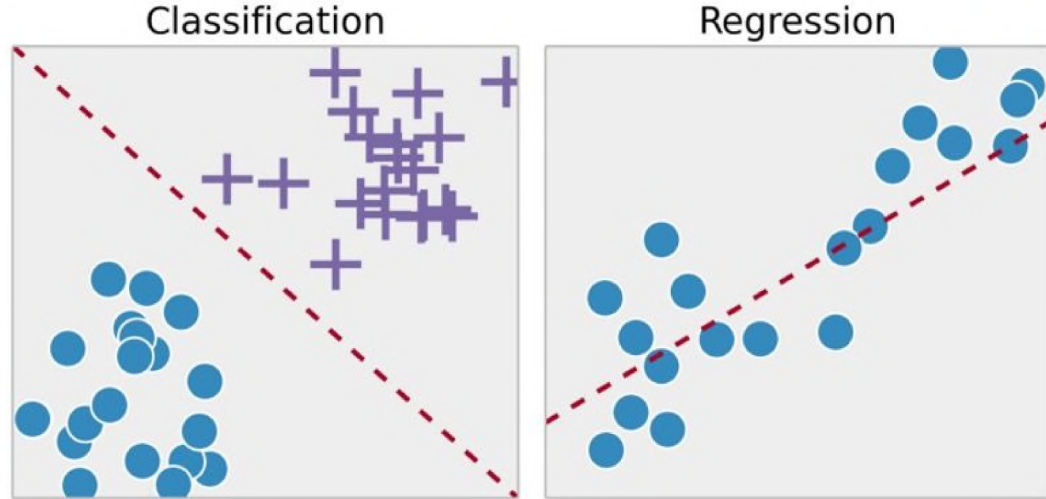
Converting from numerical to categorical - **discretization (i.e., binning)**

- Equal-width intervals: split data into ranges of the exact same size
- Equal-frequency intervals: split data so each group holds roughly same number of items
- Custom intervals: split data based on user-defined threshold



Classification vs. Regression

- **Classification** is to find the decision boundary that separates the different groups of data as clearly as possible
- **Regression** is to find the function that fits the data the best



Supervised Learning Methods

Regression

- The least-square method
- Gradient descent
- K-Nearest Neighbors (KNN), Artificial Neural Network (ANN), Naïve Bayes
- Decision tree, Random forests, Support Vector Machines (SVM), Logistic regression, etc.

Classification

- Logistic regression
- K-Nearest Neighbors (KNN)
- Artificial Neural Network (ANN), Naïve Bayes, Decision tree, Random forests, Support Vector Machines (SVM), etc.

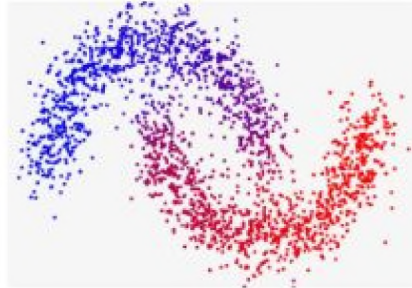
Note: Most supervised learning methods can be used for both regression and classification via small changes

Unsupervised Learning - Definition

- Useful when **NO labeled data available**
 - However, if labeled data is available → better to use supervised learning

Given any **dataset** $\{X_1, \dots, X_k\}$

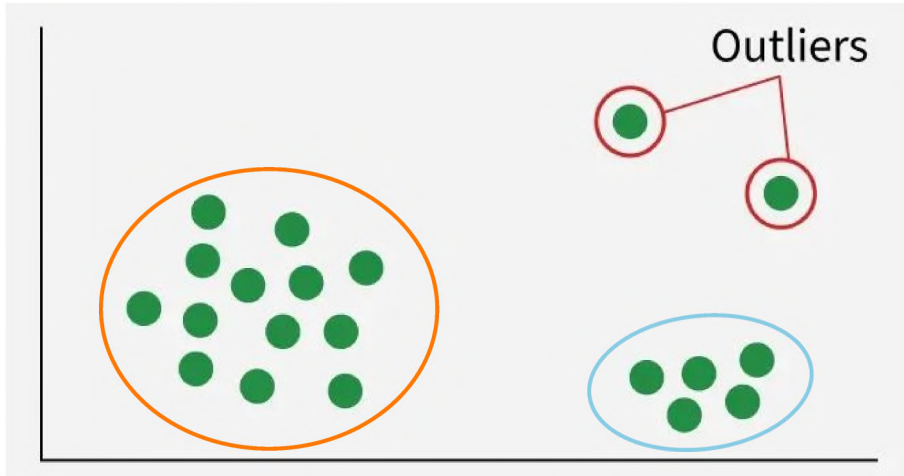
- Predict outcomes for a given X (new data):
 - Clusters instances with similar properties: **Clustering**
 - Associations instances with high occurrences together: **Association rule mining**
 - Dependencies instances with high correlations or variations: **Feature selection**



TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Unsupervised Learning - Applications

- Identifying fake news
- Object detection, recommendation systems
- Identifying fraudulent or criminal activities
- Visualization



MARKET BASKET ANALYSIS

Discovering products that are often purchased together



GENETIC CLUSTERING

Grouping genes with similar expression patterns



SOCIAL NETWORK ANALYSIS

Identifying communities within large networks



ANOMALY DETECTION

Spotting fraudulent transactions in banking

Note: In *some applications* we are interested in discovering **outliers**, not clusters (*outlier analysis*)

Unsupervised Learning - Applications (cont'd)

Feature Selection

- the process of choosing the most useful and relevant input attributes from a dataset to build an efficient machine learning model

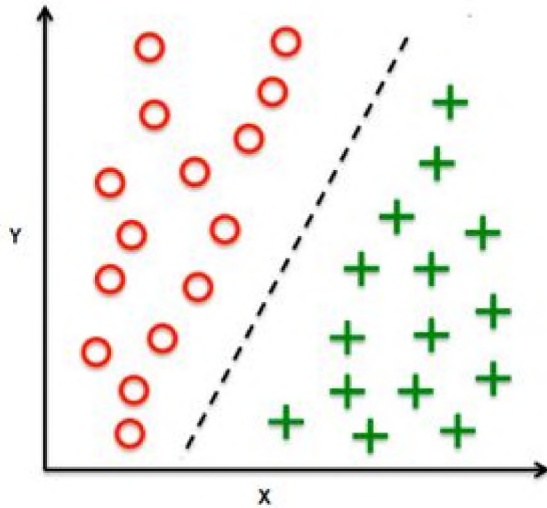
Do we need all these features?

Input	nchars	lastchar	nvowels
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Carlos	6	's'	2
Jose	4	'e'	2
Maria	5	'a'	3
Belen	5	'n'	2

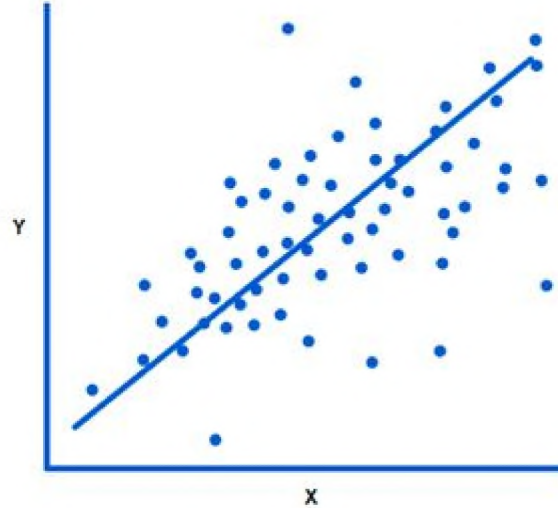
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7.0	3.2	4.7	1.4
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...	...	...	...

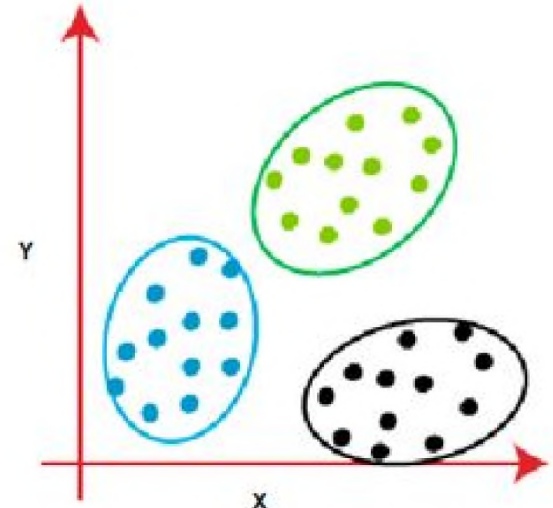
Comparison



Classification



Regression



Clustering

Unsupervised Learning - Association Rule Mining

Given a set of transactions, find **association rules** that will predict the occurrence of an item based on the occurrence of other items in a transaction

- **Association rule:** $X \rightarrow Y$
 - where X and Y are itemsets
 - e.g., {Bread, Milk} $\rightarrow$ {Diaper}

Rule evaluation metrics (strength of cause-effect)

- Support (s): Fraction of transactions that contain X U Y
 - e.g., $s = (\text{Bread, Milk, Diaper}) / |T| = 2/5 = 0.4$
- Confidence (c): Measures how often items in Y appear in transactions that contain X U Y
 - e.g., $c = \sigma(\text{Bread, Milk, Diaper}) / \sigma(\text{Bread, Milk}) = 2/3 = 0.67$

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Useful in discovering interesting relations between variables in large databases

Unsupervised Learning Methods

Clustering

- Partitional clustering
- Hierarchical clustering
- Density-based clustering, model-based clustering

Association Rule Mining

- Apriori algorithm, FP growth algorithm

Feature Selection

- Correlation/covariance matrix (to identify multicollinearity)
- Information gain (the degree of uncertainty)
- Fisher's linear discriminant analysis (LDA) (variance of the score)
- Principal Component Analysis (PCA)